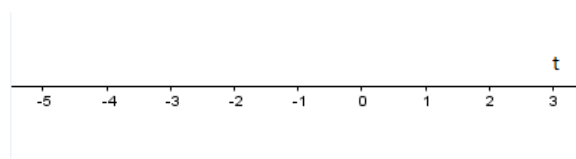


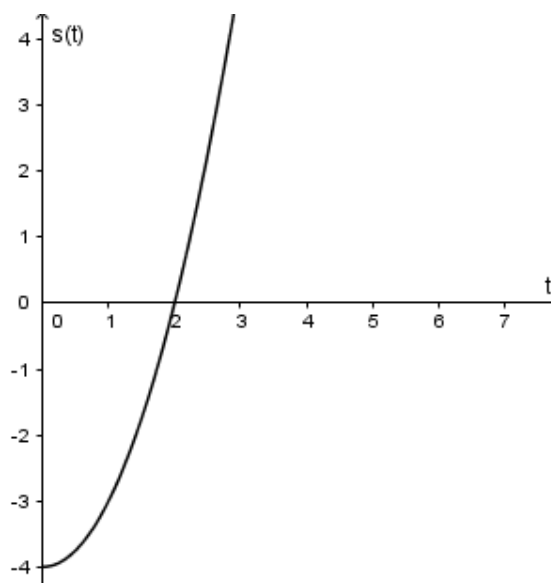
Problem 1

Problem 1 The position, $s(t)$, of an object moving along a horizontal line is given by $s(t) = t^2 - 4$, where s is in meters and t is in seconds, $0 \leq t < 5$.

- (a) Mark the position of the object on the line at time $t = 1$:



- (b) Find the average velocity, v_{AV} , of the object during the time interval $[1, 3]$.
- (c) Compute the average velocity, $v_{AV}(t)$, of the object during the time interval
- (i) $[1, t]$, for $1 < t < 5$;
 - (ii) $[t, 1]$, for $0 \leq t < 1$.
- (d) Find the instantaneous velocity, v_{inst} , of the object at $t = 1$. Justify your answer.
- (e) The position-time graph of the function s is given in the figure below.



Problem 1

- (i) Assume P is a point on the graph of s . Fill in the blank.

$$P = (1, \underline{\hspace{1cm}}).$$

- (ii) Plot the point P and draw the tangent line at this point in the figure above.
- (iii) Find the slope, m_{tan} , of the tangent line in part (ii). Explain.